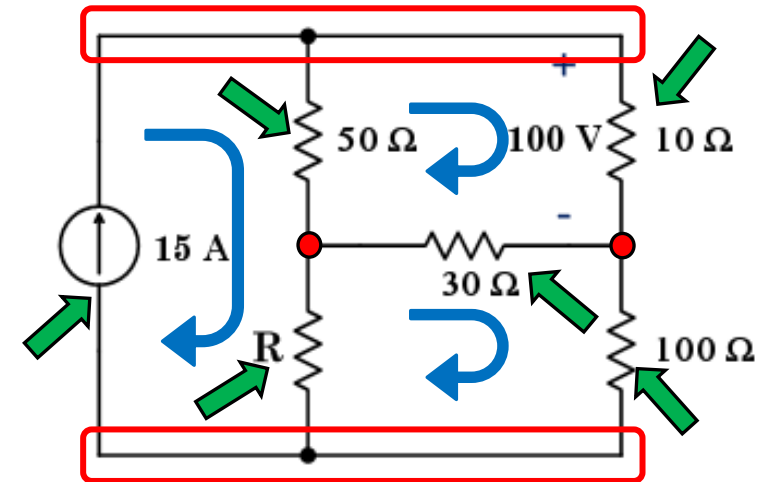
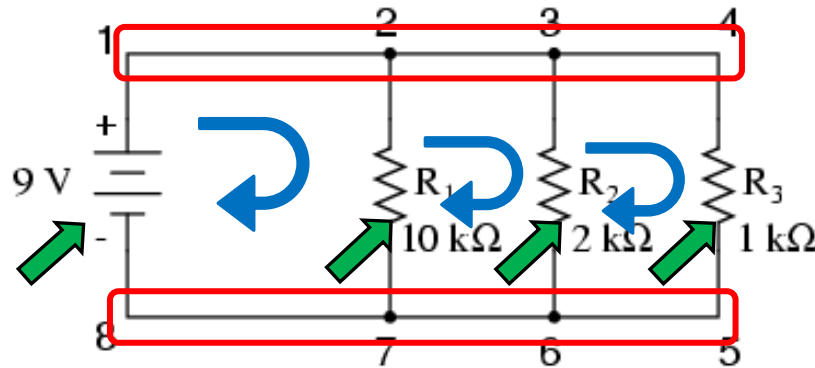
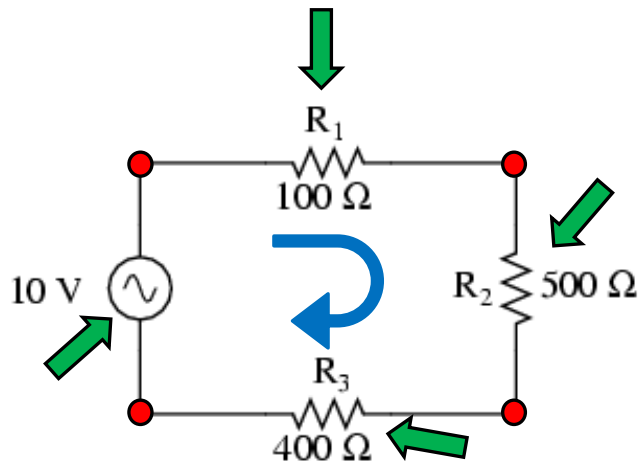




The logo of Habib University is a circular emblem. It features a purple outer ring with the text "HABIB UNIVERSITY" in white capital letters. Inside the ring is a gold-colored circle containing a profile of a lion's head facing left. Below the lion's head is a purple ribbon banner with the year "2014" written in gold.

BRANCHES, NODES AND LOOPS



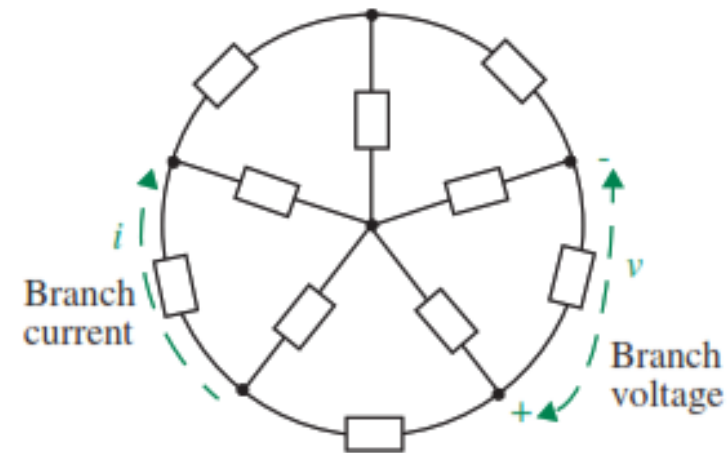
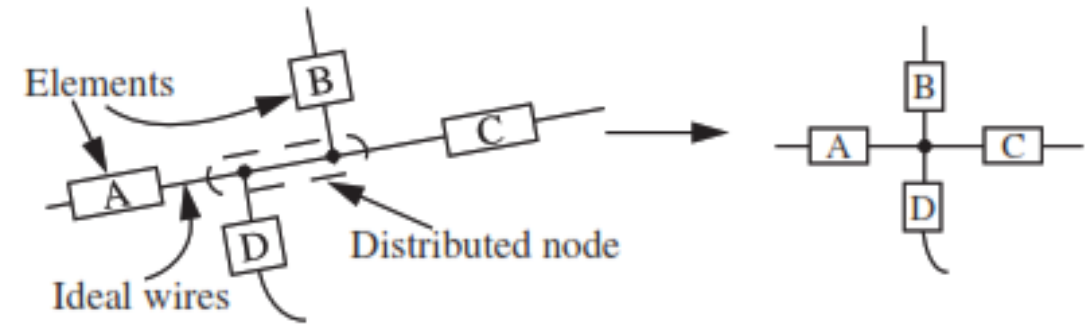
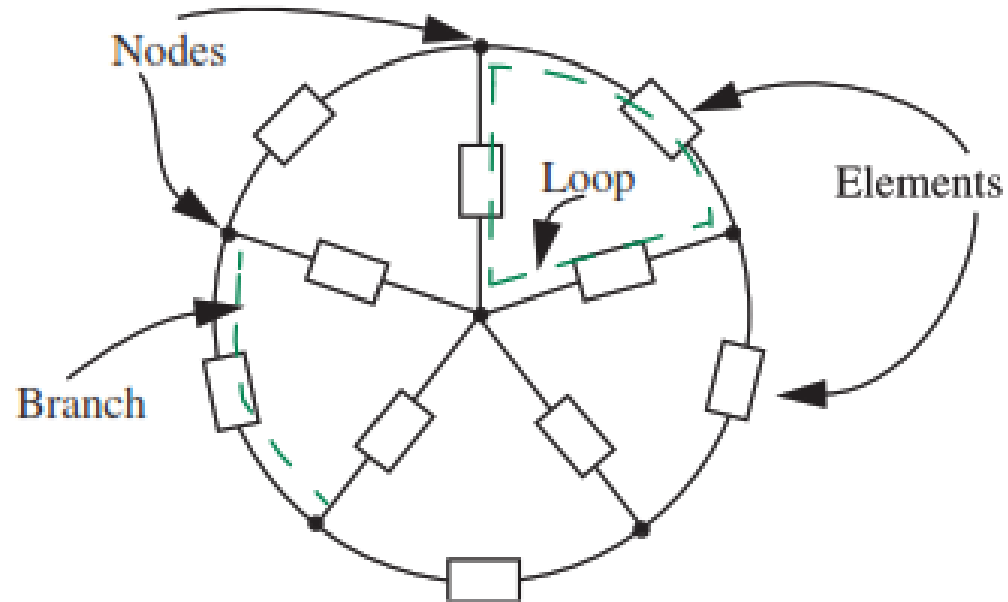
Every circuit has **branches**, **nodes** and **loops**.

A **branch** represents a single element such as a voltage source or a resistor.

A **node** is the point of connection between two or more branches.

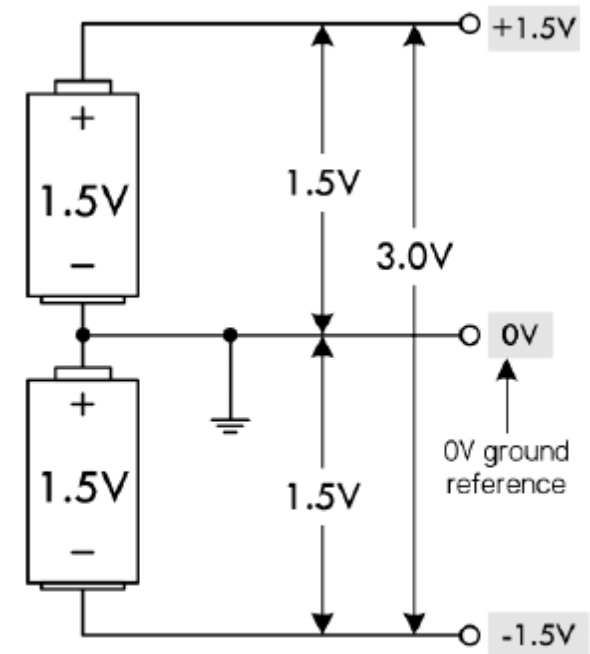
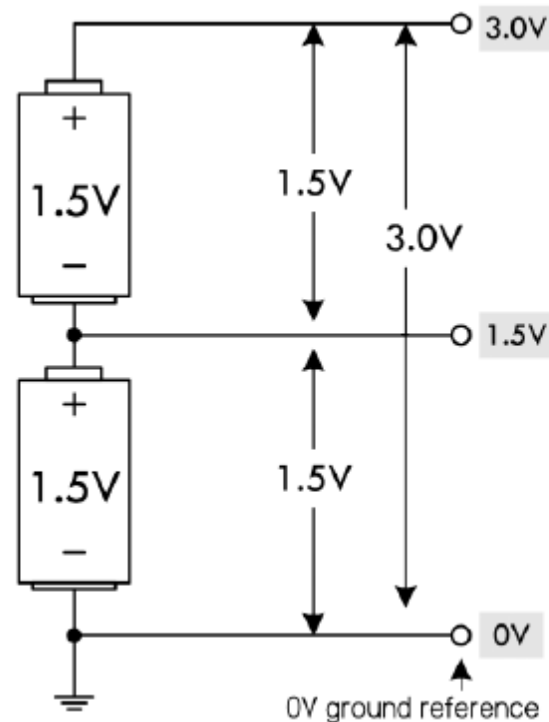
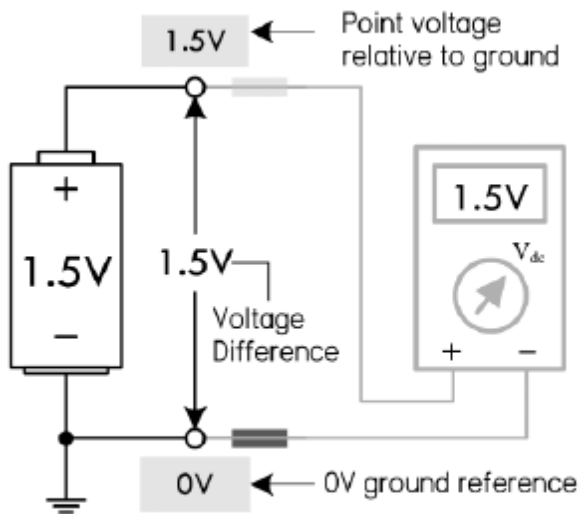
A loop is any closed path in a circuit. The smallest loop is called a **mesh**.

BRANCHES, NODES AND LOOPS

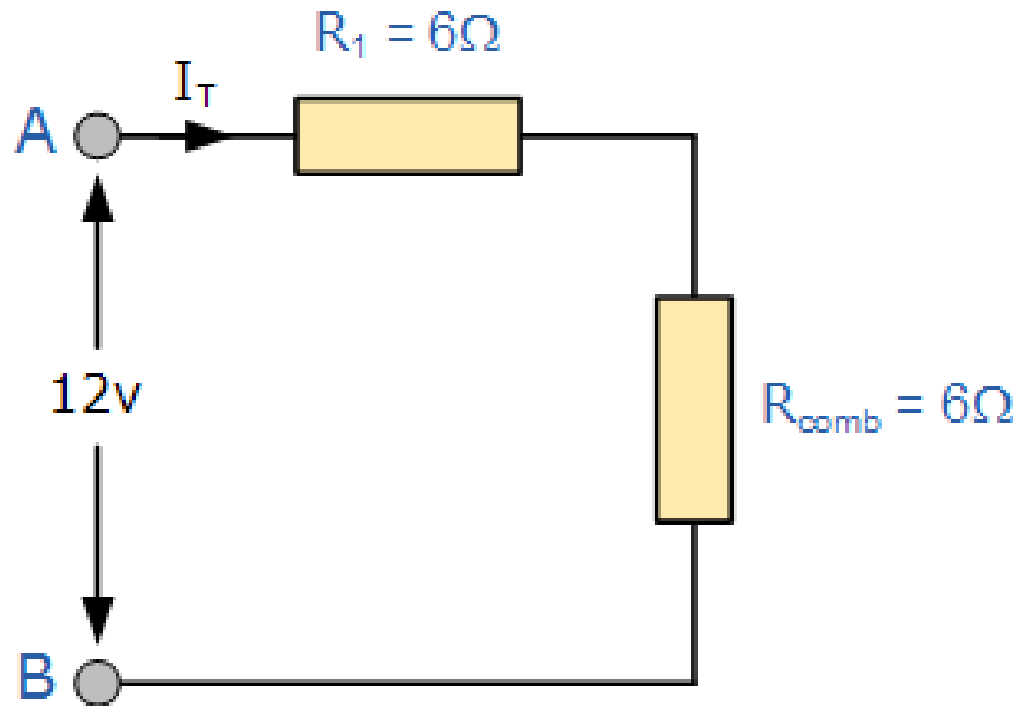


GROUND

In electric circuits, ground is a zero volt **reference point** on which to base all other voltage measurements.



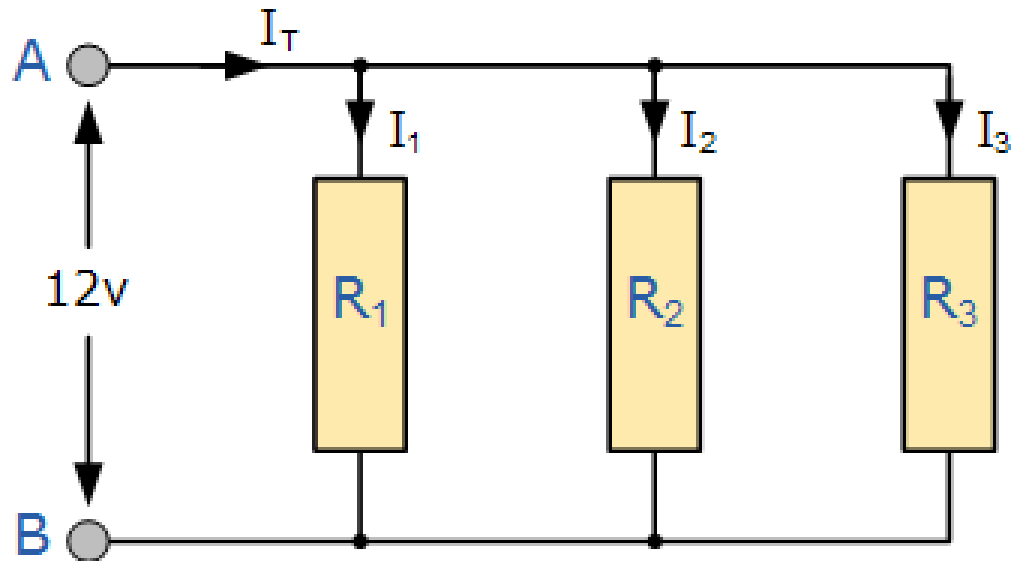
RESISTORS IN SERIES



$$R_{eq} = R_1 + R_2 + R_3 + \dots + R_n$$

- The defining characteristic of a series circuit is that there is only one path for electron flow.

RESISTORS IN PARALLEL



$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$$

- Defining characteristic is that all components are connected between the same set of nodes.

RESISTOR COMBINATIONS

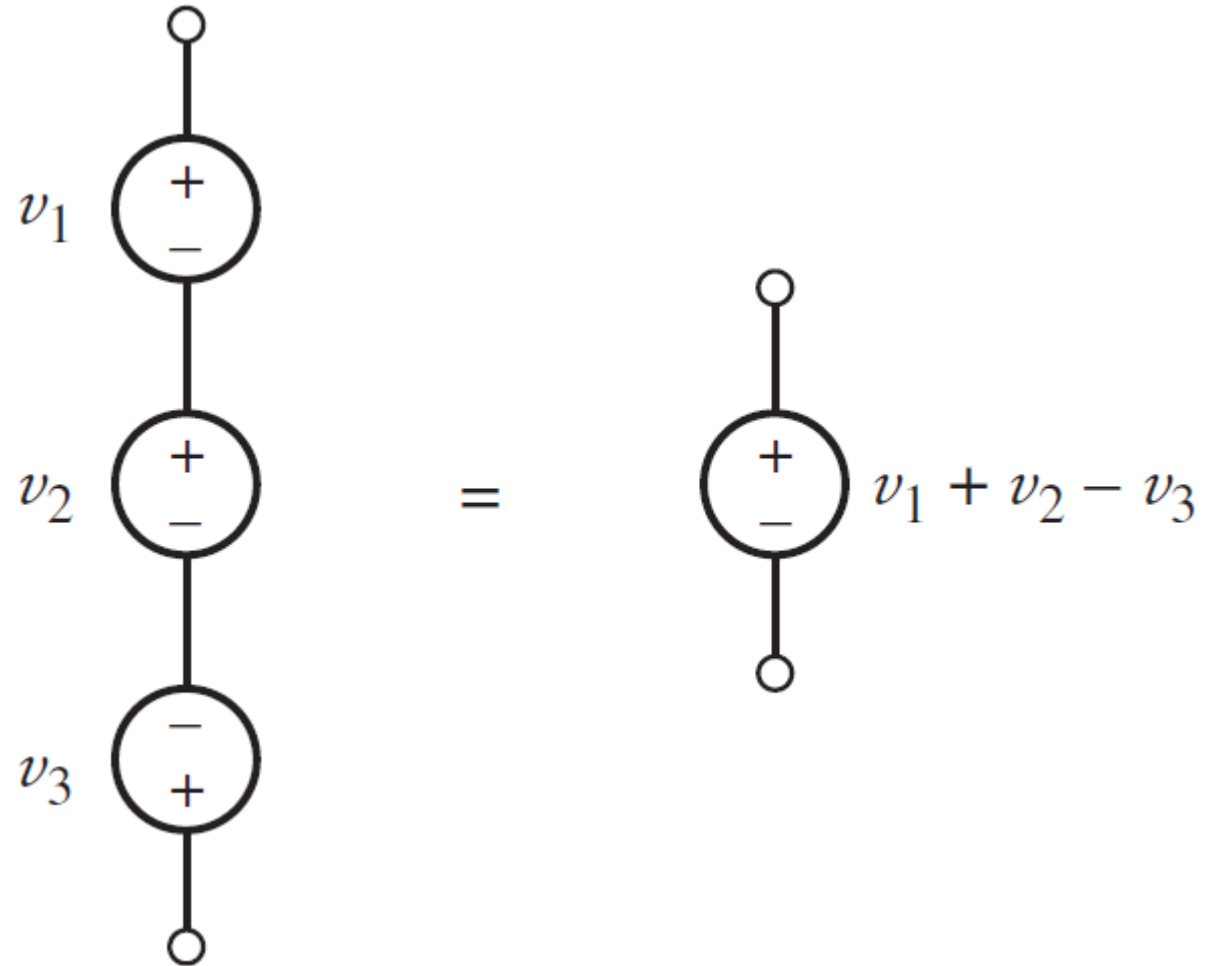
Resistors in series: $R_{eq} = R_1 + R_2 + R_3 + \dots + R_n$

- R_{eq} will always be greater than the highest resistance in that series.
- If $R_1, R_2, R_3, \dots R_n = R$, then $R_{eq} = nR$.
- Two series resistors share one node.

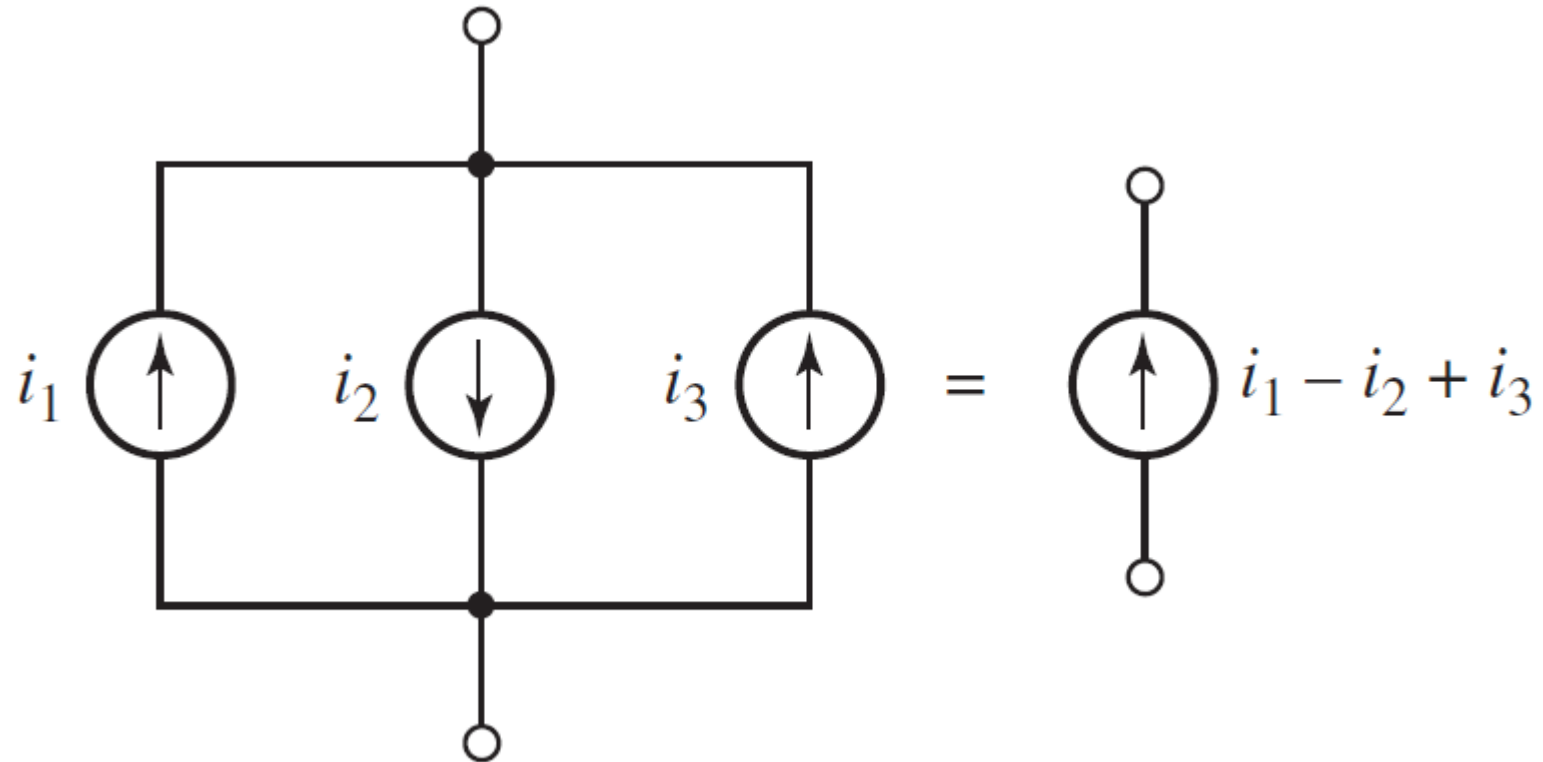
Resistors in parallel: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$

- R_{eq} will always be lower than the lowest resistance in parallel.
- If $R_1, R_2, R_3, \dots R_n = R$, then $R_{eq} = R/n$.
- Two parallel resistors share two nodes.

SOURCE COMBINATIONS



SOURCE COMBINATIONS



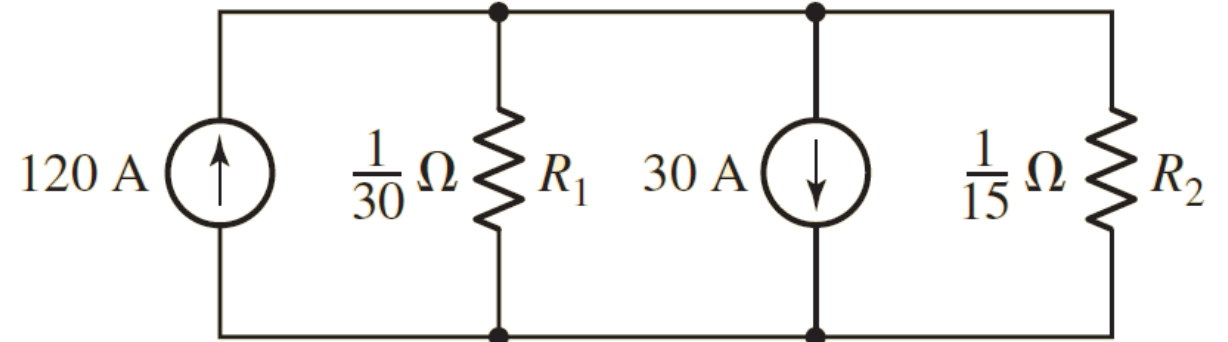
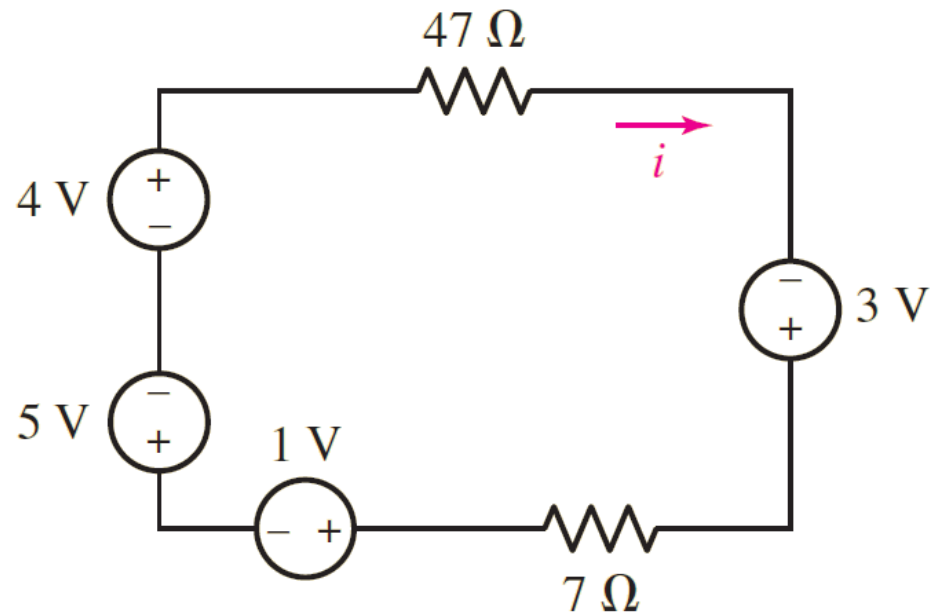
SOURCE COMBINATIONS

Steps for combining sources:

1. Choose a direction of your final combined source.
2. Add all sources in the direction you chose and subtract those in the opposite direction.
3. If the result is positive, draw the chosen direction. If it is negative, draw in the opposite direction.

PRACTICE

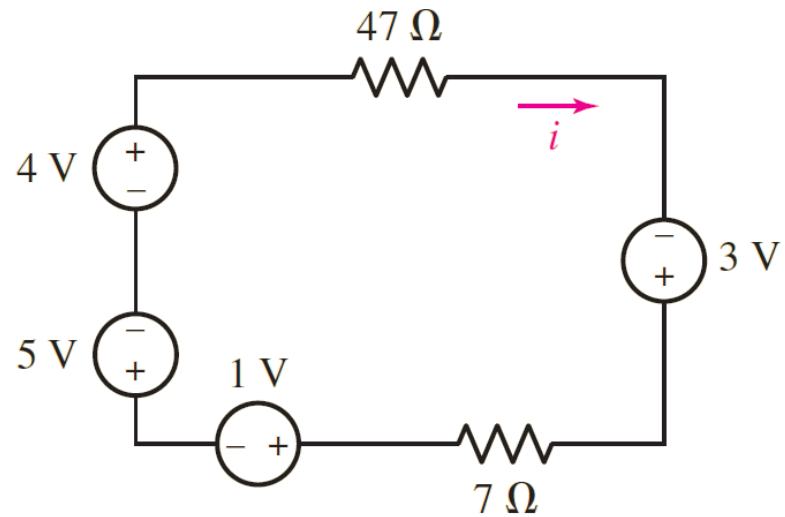
Simplify the circuits below:



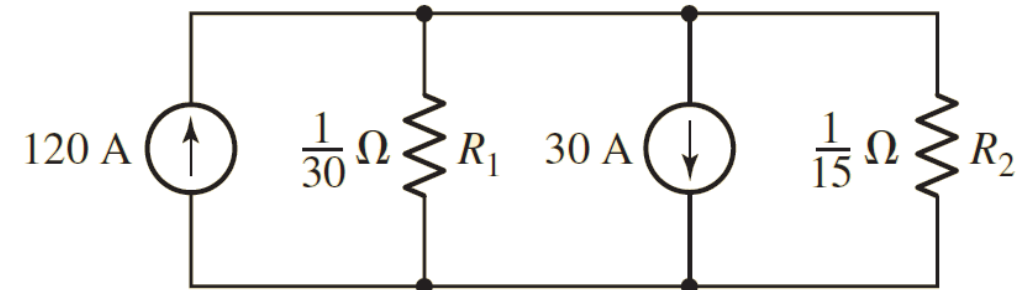
Practice:



Simplify the circuits below:



$$R = 47 + 7$$

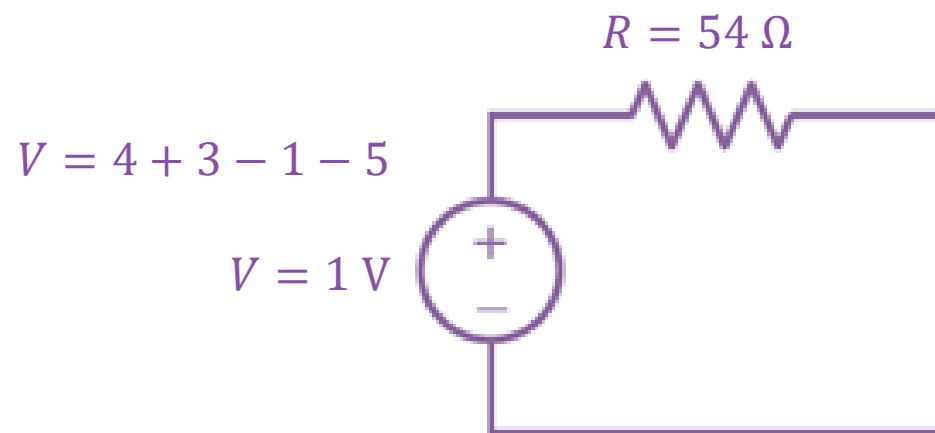


$$I = 120 - 30$$

$$I = 90 \text{ A}$$

$$R = \frac{1}{\frac{1}{30} + \frac{1}{15}}$$

$$R = \frac{1}{45} \Omega$$



ANALYZE OUR FIRST CIRCUIT

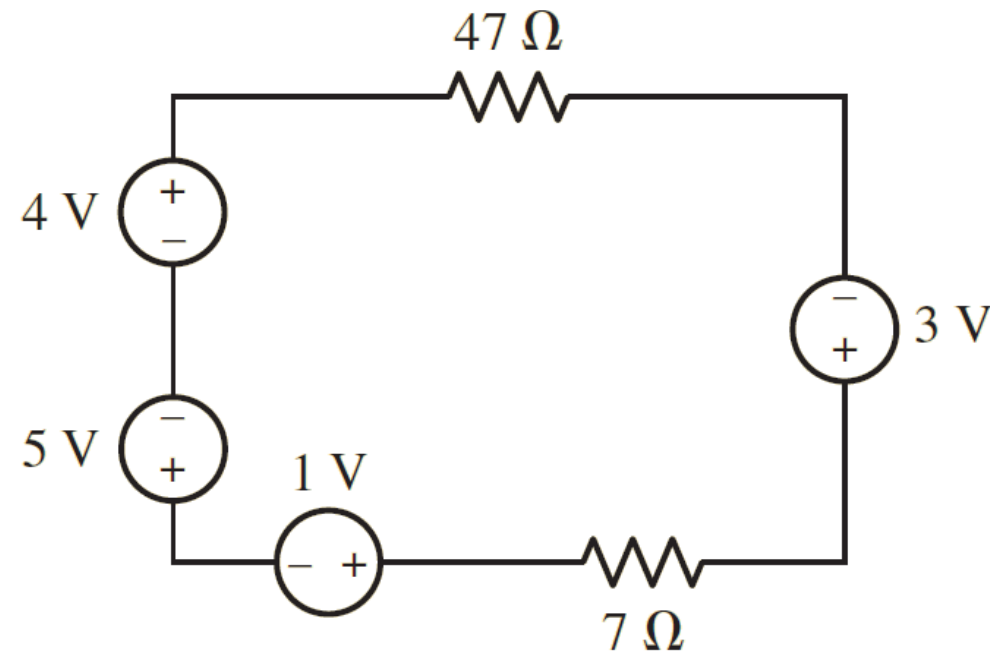
Exercise: What is the voltage across each resistor?

Hint: Calculate the total voltage, the total current, and then use Ohm's Law for each resistor.

Answers:

$$i = 18.5 \text{ mA}$$

$$v_{47\Omega} = 870 \text{ mV}, \quad v_{7\Omega} = 130 \text{ mV}$$

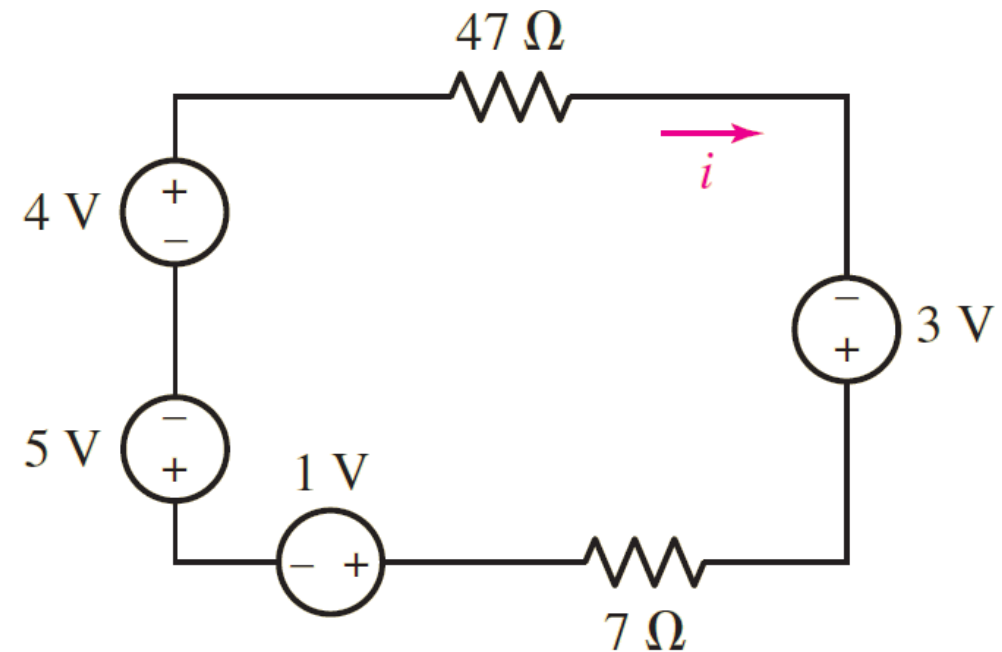


RESISTORS IN SERIES

If you add the individual voltages, you will find that it equals to the total voltage supply.

So we have learnt two things from this exercise for resistors in series:

1. The current remains the same.
2. The voltage supplied is totally divided among them.



KIRCHHOFF'S VOLTAGE LAW

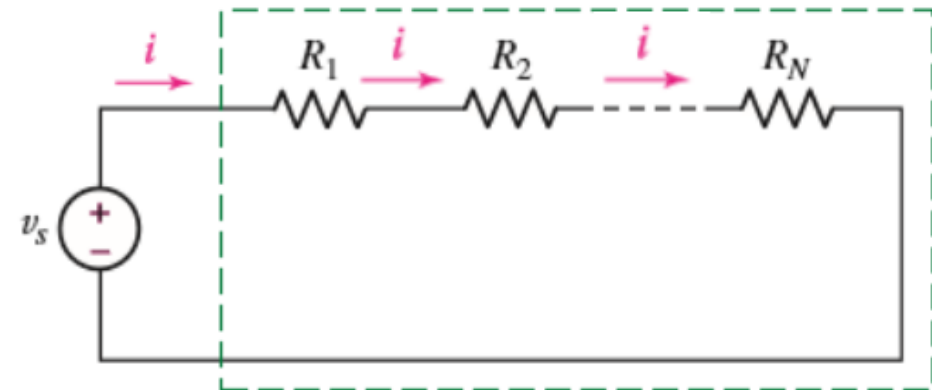
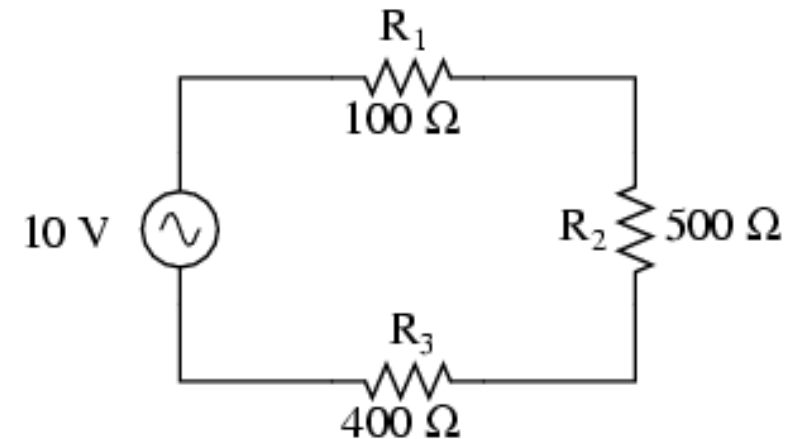
The algebraic sum of the voltages around any loop of a circuit is zero.

$$V_1 + V_2 + V_3 + \dots + V_n = 0 \text{ or } \sum_{n=1}^N V_n = 0$$

Can we be sure that this will hold for all circuits?

Law of conservation of energy:

The total voltage or power supplied is equal to the total voltage or power dropped.



KIRCHHOFF'S VOLTAGE LAW

Another perspective: Let's consider the power being supplied by the voltage source ($P = VI$).

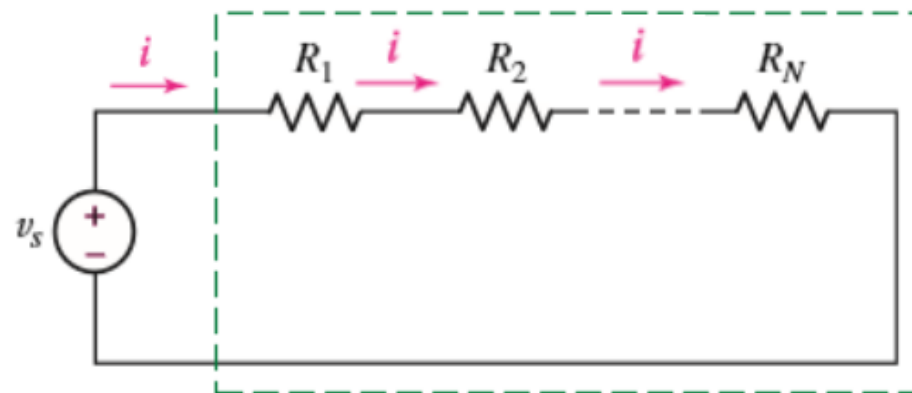
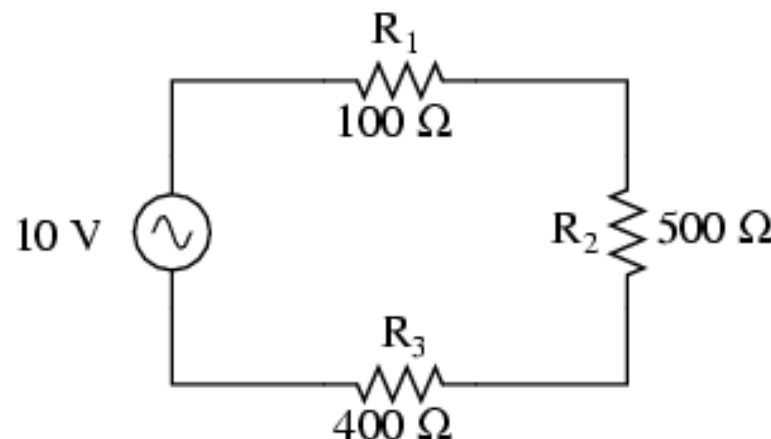
The individual power dissipated by each resistor would be $P_1 = V_1I$, $P_2 = V_2I$ and $P_3 = V_3I$.

By Law of Conservation of Energy:

$$P = P_1 + P_2 + P_3$$

$$VI = V_1I + V_2I + V_3I$$

$$V = V_1 + V_2 + V_3$$



WRITING KVL EQUATIONS FOR A LOOP

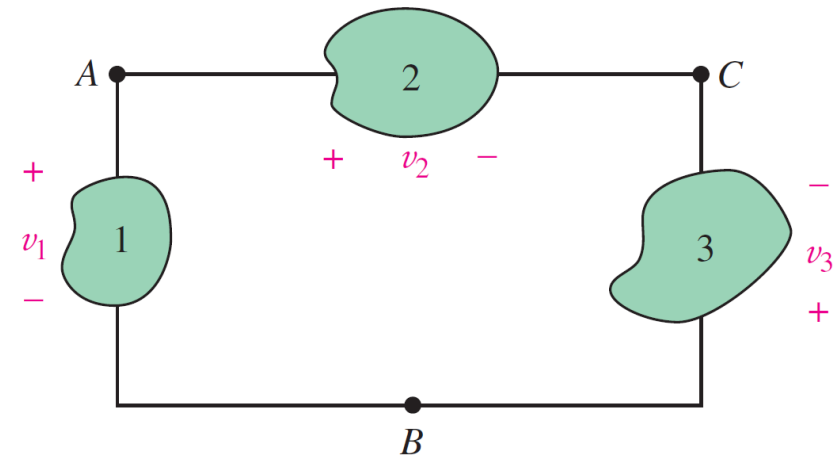
Consider current direction clockwise, and write KVL equation by adding all voltages on each element in a loop or mesh.

If current enters from positive terminal, write voltage sign as positive and vice versa.

Applying this to the single loop:

$$-v_1 + v_2 - v_3 = 0$$

Useful when circuit simplification is too lengthy or impossible.

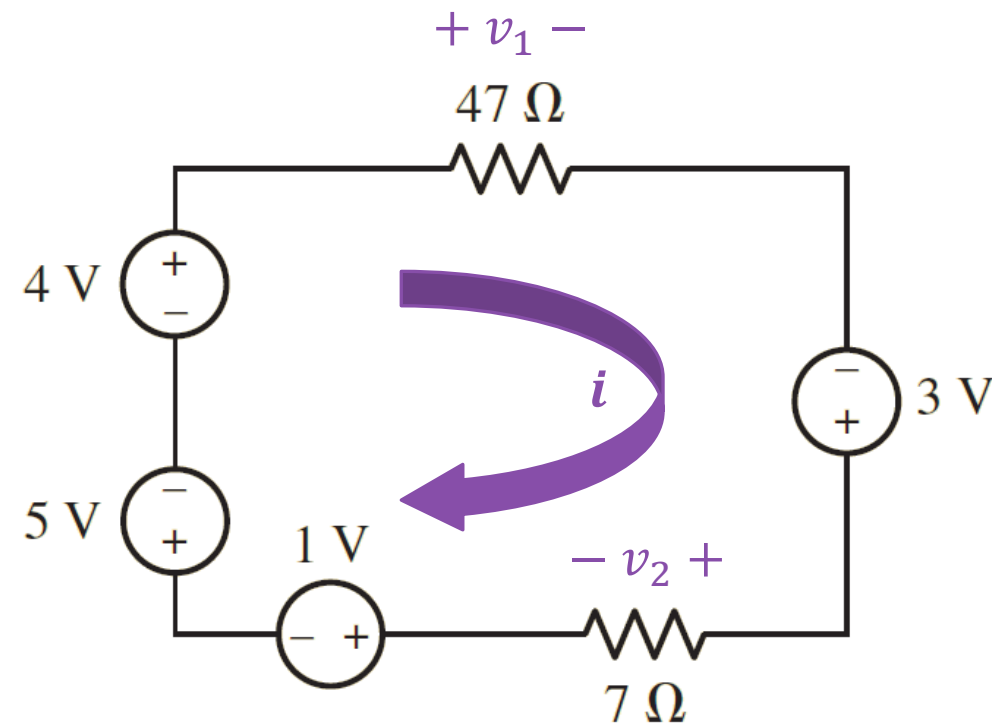


SINGLE LOOP ANALYSIS

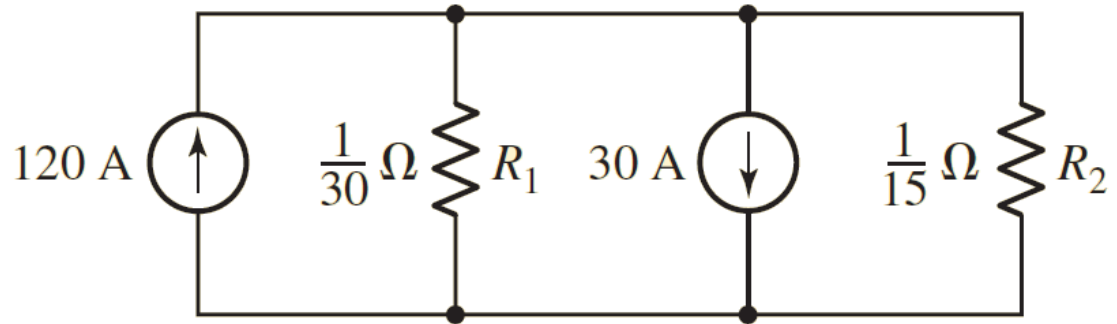
1. Identify the mesh in the circuit and assign reference direction of current in the loop.
2. Using associated variables convention, assign voltage polarity to each element that is not assigned one.
3. Apply KVL on loop having current i . Use Ohm's Law for unknown values:

$$5 - 4 + 47i - 3 + 7i + 1 = 0$$

4. Solve the equation to attain the total current, with which you can find individual voltages across the resistors.



ANALYZE OUR SECOND CIRCUIT



Exercise: Find the current through each resistor?

Hint: Calculate the voltage across the resistors after circuit simplification and use Ohm's Law.

Answers:

$$v = 2 \text{ V}$$

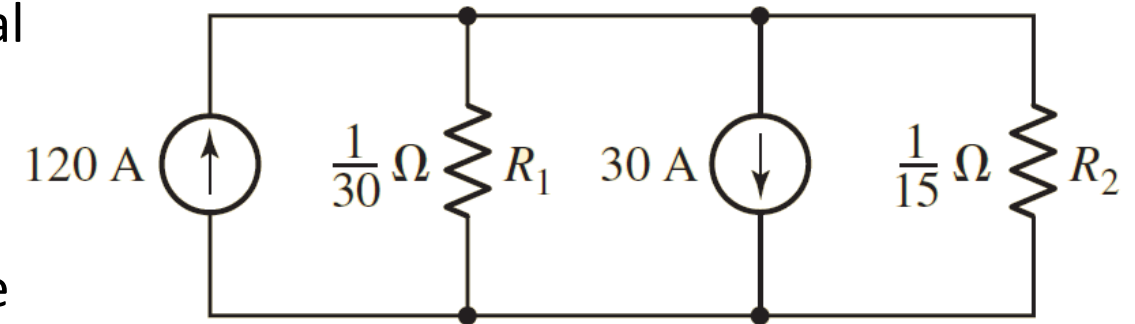
$$i_{1/30\Omega} = 60 \text{ A}, i_{1/15\Omega} = 30 \text{ A}$$

RESISTORS IN PARALLEL

You will find that all currents add up to the total value that is being supplied by the sources.

So we have learnt two things from this exercise for resistors in parallel:

1. The voltage remains the same.
2. The current supplied is divided among the branches, or the sum of the individual currents is equal to the total current supplied.



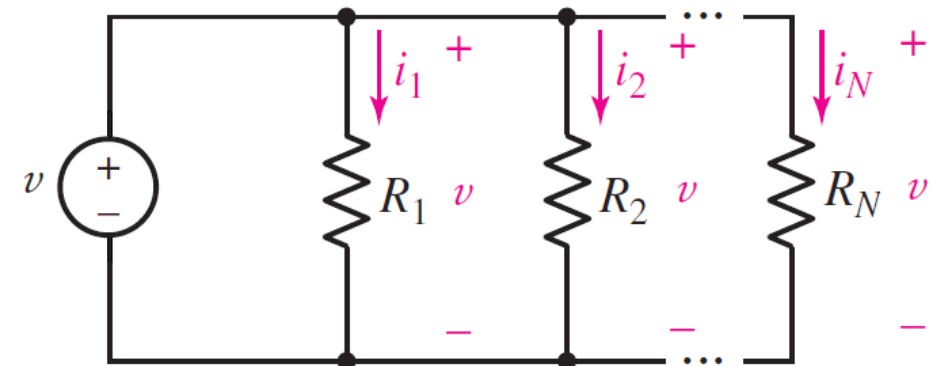
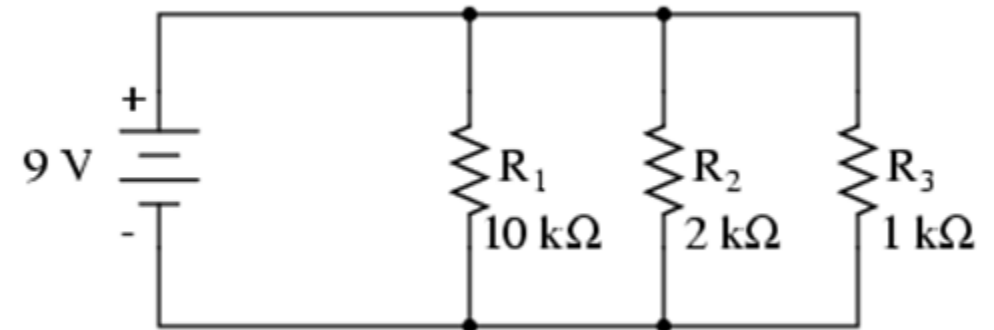
KIRCHHOFF'S CURRENT LAW

The algebraic sum of the currents entering or exiting a node of a circuit is zero.

$$i_1 + i_2 + i_3 + \dots + i_n = 0 \text{ or } \sum_{n=1}^N i_n = 0$$

Can we relate this to a conservation law as well?

Law of conservation of energy:
Charges can neither be created, nor destroyed.



KIRCHHOFF'S CURRENT LAW

Another perspective: Let's consider the power being supplied by the voltage source ($P = VI$).

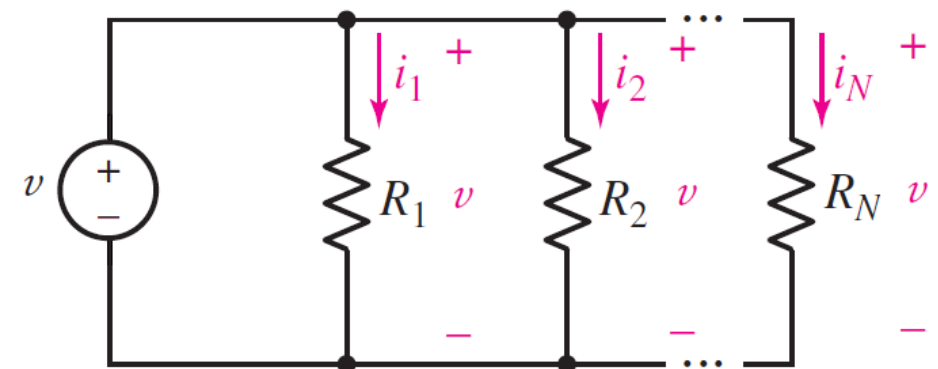
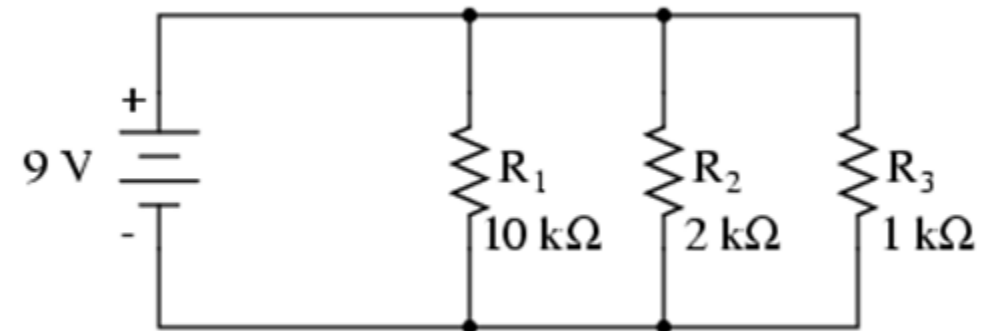
The individual power dissipated by each resistor would be $P_1 = VI_1$, $P_2 = VI_2$ and $P_3 = VI_3$.

By Law of Conservation of Energy:

$$P = P_1 + P_2 + P_3$$

$$VI = VI_1 + VI_2 + VI_3$$

$$I = I_1 + I_2 + I_3$$



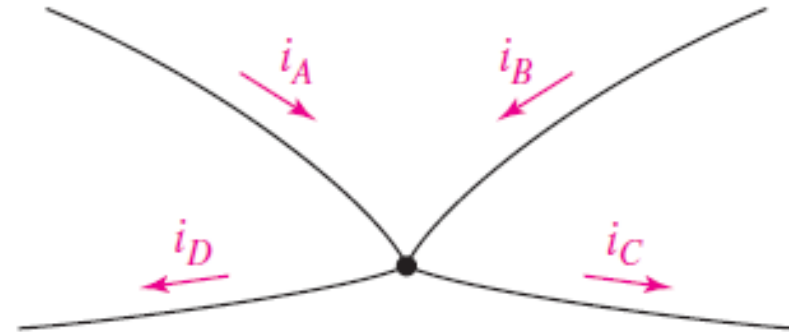
WRITING KCL EQUATIONS FOR A NODE

Consider current exiting a node as positive.

Write KCL equation by adding all currents through each element connected to a node.

Applying this to the single node:

$$-i_A - i_B + i_C + i_D = 0$$

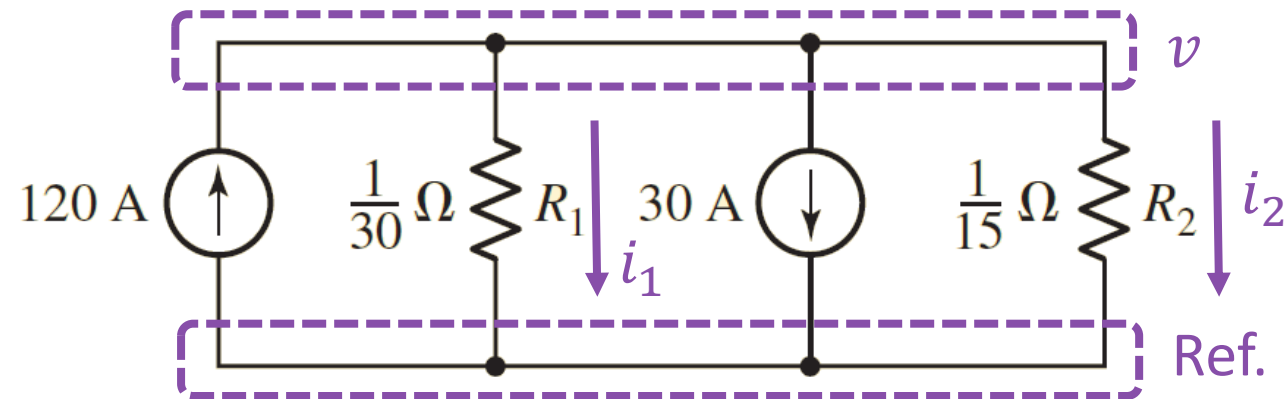


SINGLE NODE-PAIR ANALYSIS

1. Identify the nodes in the circuit, and assign voltage v to any node with respect to reference node.
2. Assign current direction with respect to the associated variables convention.
3. Apply KCL on node having voltage v . Use Ohm's Law for unknowns:

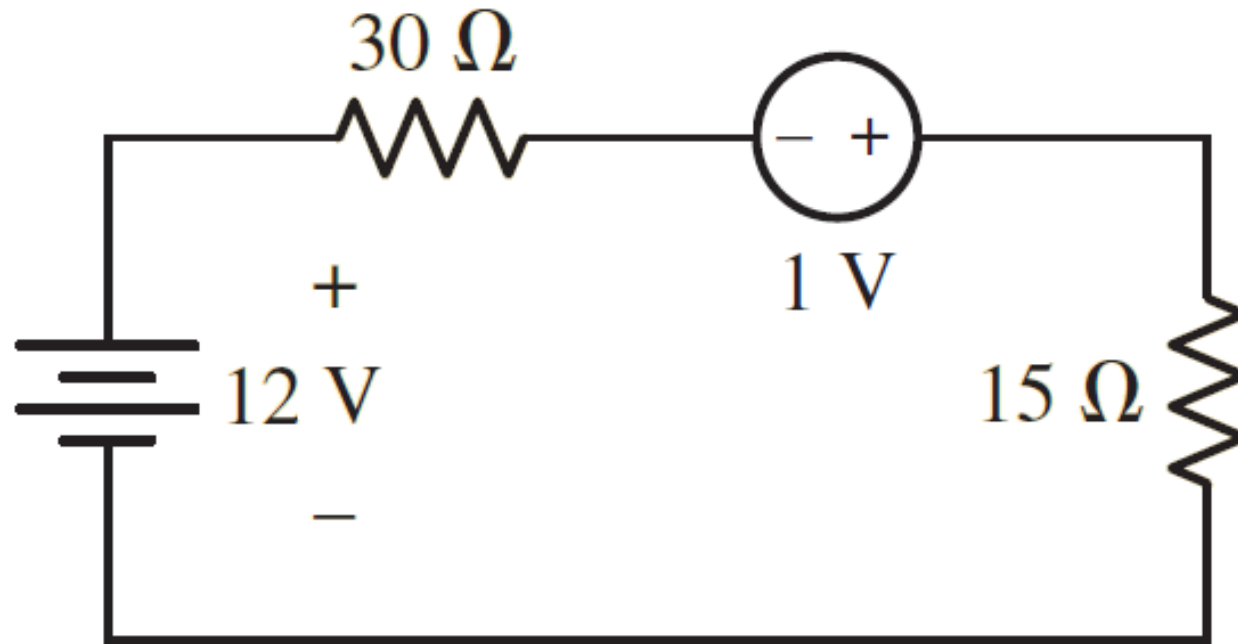
$$-120 + \frac{v}{1/30} + 30 + \frac{v}{1/15} = 0$$

4. Solve the equation to attain the total voltage, with which you can find individual currents through the resistors.



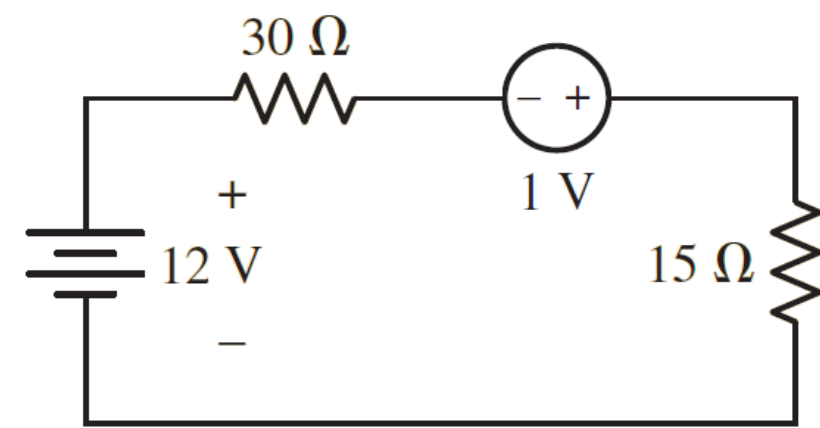
PRACTICE

Compute the total current and the voltage drop of the individual resistors.



Practice:

Compute the total current and the voltage drop of the individual resistors.



KVL equation for this loop:

$$-12 + 30i - 1 + 15i = 0$$

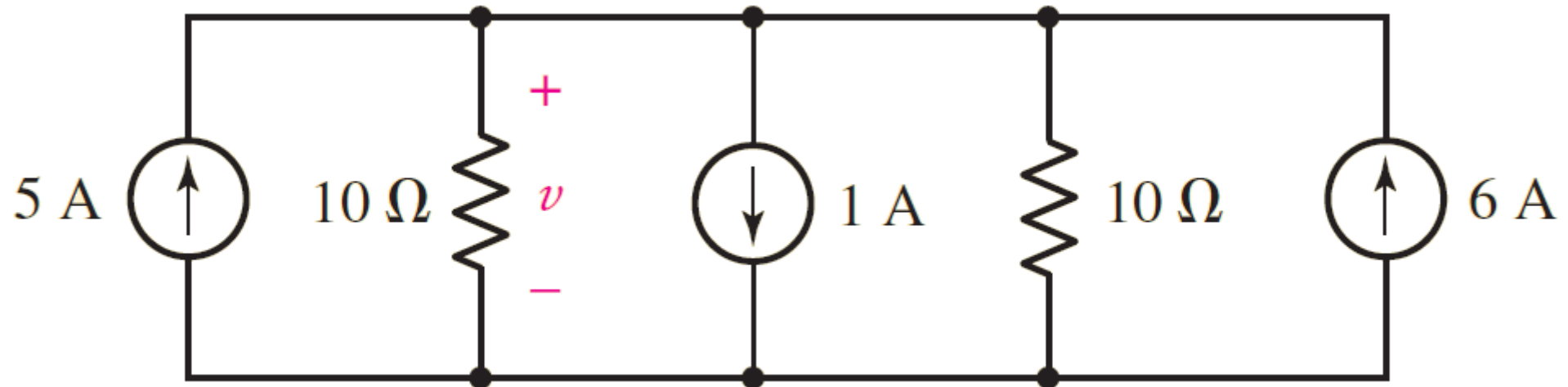
$$i = 289 \text{ mA}$$

$$v_{30\Omega} = 0.289 \times 30 = 8.67 \text{ V}$$

$$v_{15\Omega} = 0.289 \times 15 = 4.33 \text{ V}$$

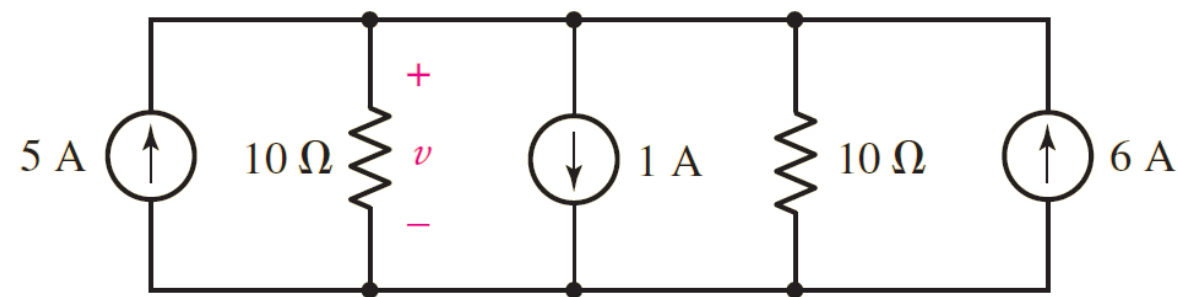
PRACTICE

Calculate v and all unknown branch currents for the given circuit.



Practice:

Calculate v and all unknown branch currents for the given circuit.



Applying KCL:

$$-5 + \frac{v}{10} + 1 + \frac{v}{10} - 6 = 0$$

$$v = 50 \text{ V}$$

$$i_{10 \Omega} = \frac{50}{10} = 5 \text{ A}$$